Supplementary Material for: GroundSLAM: A Robust Visual SLAM System for Warehouse Robots Using Ground Textures

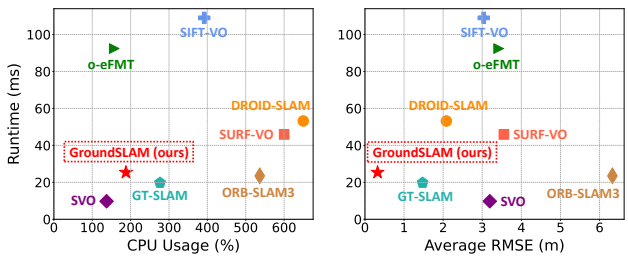

Fig. 1: Efficiency analysis for different systems on the GeoTracking dataset. The vertical axis is the average runtime of each frame. The horizontal axes are the CPU usage (left) and average RMSE (right), respectively.

1. Efficiency Analysis

This section presents the efficiency analysis of our GroundSLAM system. The evaluation is conducted on a computer with an Intel Core i9 CPU and the image resolution is 640×480. We compare the running speed, computing resource consumption, and the accuracy of different systems. The metrics are per-frame runtime, CPU usage, and average RMSE. For sequences where tracking failures occur, we assign an error of 10 meters when computing the average RMSE. The results are presented in Fig. 1. Overall, our system achieves the highest accuracy while maintaining real-time performance with minimal computational resource consumption. GroundSLAM demonstrates a runtime comparable to GT-SLAM and ORB-SLAM3 while utilizing fewer CPU resources. Although o-eFMT is also a correction-based ground-texture localization system, it exhibits significantly lower efficiency and accuracy compared to our approach, further validating the effectiveness of the proposed methods. It is also worth noting that DROID-SLAM actually utilizes more than 6 CPU cores, and its CPU usage in Fig. 1 is only for a compact presentation.

2. Live Mapping Demo

We present a real-time mapping demonstration incorporating loop detection and correction. The data was collected in a warehouse environment using an Automated Guided Vehicle (AGV). As shown in Fig. 2, the robot followed a rectangular trajectory (highlighted in green), starting and stopping at the same location (marked by the red square), thereby forming a loop. Images along this path were then stitched using poses estimated by GroundSLAM, both without (left) and with (right) loop correction. Note that the tag was not utilized for pose estimation but served solely as a reference ground truth. Due to drift errors, the stitched map without loop correction exhibits noticeable blurring, which is effectively eliminated on the right. This clarity underscores the effective elimination of drift errors and the precise pose corrections. Additionally, we have tested GroundSLAM with this AGV in a real industrial environment spanning over 10,000 m^2. The system exhibited consistent robustness and accuracy. Refer to `https://youtu.be/2sI76SDptDA` for more details.

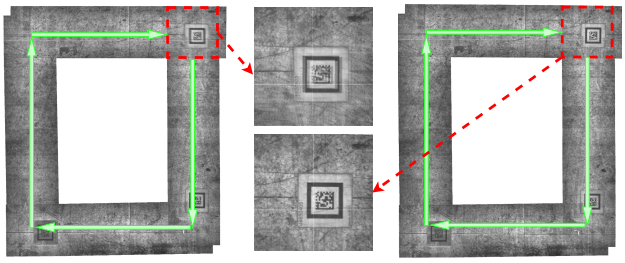

Fig. 2: A live mapping demo of our system without (left) and with (right) loop correction. The robot moves in a rectangular path (the green line) with the starting and ending being at the same location (the red square). The drift error of our VO module is only **0.2%** of the trajectory, which is significantly lower than other SOTA systems (about 1%). The drift error causes the blurring in the stitched map (left), and it is eliminated by our loop correction (right).

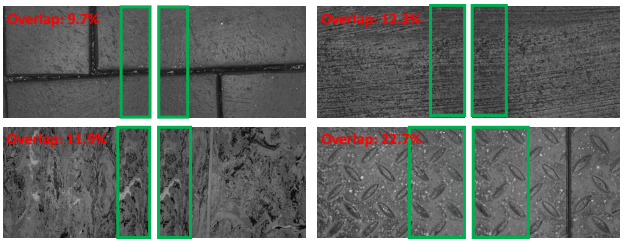

Fig. 3: Some failure cases of our system on the HD Ground dataset. As the publicly available sequences are sampled from the original captured sequences, the overlaps of some adjacent images in this dataset are very small.

3. Failure Case Analysis

We observed that many failure cases of our method stem from the low image overlap in the HD Ground dataset, as shown in Fig. 3. To further analyze the effect of overlap rate on performance, we uniformly sample the original PathTex sequences to create new sequences with varying overlaps. Fig. 4 shows the average overlap rates and localization errors on the "Coarse_asphalt" texture under different sampling intervals. Two inflection points are observed: when the

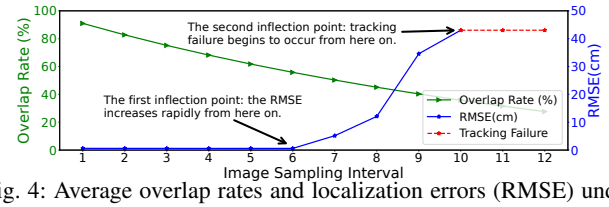

Fig. 4: Average overlap rates and localization errors (RMSE) under different sampling intervals on the “Coarse_asphalt" texture. When the overlap rate is less than 50%, the localization error begins to increase rapidly. When the overlap rate is less than 32%, the tracking failure begins to appear.

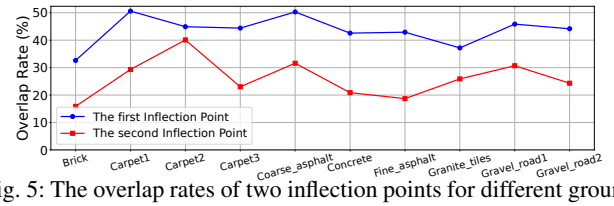

Fig. 5: The overlap rates of two inflection points for different ground textures. The average overlap rates of the first inflection point and the second inflection point are about 43% and 25%, respectively.

overlap rate drops below 50%, localization error increases rapidly, and when it falls below 32%, tracking failures begin to occur. Similar trends are observed across other textures. Fig. 5 summarizes the overlap rates at these inflection points for different ground textures, showing slight variations across textures. On average, the first and second inflection points occur at 43% and 25% overlap, respectively. Based on these findings, we conclude that for high-precision performance, GroundSLAM requires an overlap rate greater than 43% between adjacent images, while maintaining an overlap above 25% is necessary to prevent tracking failures.

In our real-world application, our warehouse robot operates at a maximum speed of 2.5m/s, ensuring a minimum image overlap of 53%. As a result, our system can provide reliable localization for our warehouse robots. To deploy our system on a faster-moving robot, a higher-frame-rate camera or a higher-mounted camera with a wider field of view can be used to ensure sufficient overlap between adjacent frames.